

IEOR 4404: Simulation

Lecture 4: Acceptance-Rejection Method

Acceptance Rejection Method

A method to generate “difficult” r.v. from “easy” ones.

- We want to generate a r.v. with density $f(x)$ (difficult)
- We can generate a r.v. with density $g(x)$ (easy)

Acceptance Rejection Method

Acceptance rejection method:

- Find a constant C such that

$$\frac{f(x)}{g(x)} \leq C \text{ for all possible } x$$

- To generate one output,
 - Generate $Y \sim g(y)$
 - “Accept” Y with probability $\frac{f(Y)}{Cg(Y)}$; otherwise “reject” and go back to the beginning
- The r.v. from $g(x)$ is called the proposal r.v.

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- Find a constant C such that

$$\frac{f(x)}{g(x)} \leq C \text{ for all possible } x$$

- To generate one output,
 - Generate $Y \sim g(y)$
 - Generate $U \sim Unif(0,1)$
 - If $U \leq \frac{f(Y)}{Cg(Y)}$, set $X = Y$ (accept); otherwise, go back to the beginning (reject)
 - Output X

Example

- Generate a r.v. with pdf

$$f(x) = 3x^2, \quad 0 \leq x \leq 1$$

- Use a *Unif*(0,1) proposal

$$g(x) = 1, \quad 0 \leq x \leq 1$$

Example

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- We have

$$\frac{f(x)}{g(x)} = \frac{3x^2}{1} = 3x^2 \leq 3 \text{ for any } 0 \leq x \leq 1$$

- Thus we can choose $C = 3$

Example

Acceptance-rejection method:

- Generate $Y \sim \text{Unif}(0,1)$
- Generate $U \sim \text{Unif}(0,1)$
- If $U \leq \frac{f(Y)}{cg(Y)} = Y^2$, set $X = Y$ (accept); otherwise, go back to the beginning (reject)
- Output X

Example

- Generate the standard normal r.v.

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, \quad -\infty < x < \infty$$

- Use a proposal

$$g(x) = \frac{1}{2} e^{-|x|}, \quad -\infty < x < \infty$$

Implementation

In the first example, suppose the sequence of $\text{Unif}(0,1)$ generated by the computer is:

0.10 0.32 0.50 0.49 0.23 0.37 0.58 0.32 0.39 0.99

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The A-R algorithm runs as:

$Y=0.10$

$U=0.32$

Since $U = 0.32 > Y^2 = 0.01$, continue

$Y=0.50$

$U=0.49$

Since $U = 0.49 > Y^2 = 0.25$, continue

$Y=0.23$

$U=0.37$

Since $U = 0.37 > Y^2 = 0.0529$, continue

$Y=0.58$

$U=0.32$

Since $U = 0.32 \leq Y^2 = 0.3364$, stop

Output = 0.58

Intuition

Motivation:

- Since $f(x)$ is difficult to sample from, we want to use an easier density $g(x)$
- However, we cannot just change the problem to sampling from $g(x)$

Mechanism:

- $Cg(x)$ is an “envelope” that covers $f(x)$
- C must be at least 1; if $C = 1$, then $f \equiv g$
- The acceptance probability $f(x)/Cg(x)$ is an adjustment from sampling $g(x)$ to sampling $f(x)$

Validity

The A-R algorithm indeed generates a random variable with density $f(x)$

First note that

$$\begin{aligned} P(\text{accept}) &= \int_{-\infty}^{\infty} P(\text{accept} | Y = x) g(x) dx \\ &= \int_{-\infty}^{\infty} \frac{f(x)}{C g(x)} \times g(x) dx \\ &= \frac{\int_{-\infty}^{\infty} f(x) dx}{C} \\ &= \frac{1}{C} \end{aligned}$$

where we use the law of total probability in the first equality

Validity

Next consider

$$\begin{aligned} & P(X \in (x, x + \Delta x) | \text{accept}) \\ &= \frac{P(X \in (x, x + \Delta x), \text{accept})}{P(\text{accept})} \\ &= \frac{P(Y \in (x, x + \Delta x)) \times P(\text{accept} | Y \in (x, x + \Delta x))}{P(\text{accept})} \\ &= \frac{g(x)\Delta x \times \frac{f(x)}{Cg(x)}}{\frac{1}{C}} \\ &= f(x)\Delta x \end{aligned}$$

Efficiency

- The runtime of the algorithm depends on the number of trials to obtain one acceptance
- Since the algorithm requires a random number of trials, it makes sense to consider the expected number of trials as an efficiency measurement.
- The number of trials to get one acceptance is $\text{Geo}(p)$, where p is the acceptance probability
- The acceptance probability is $p=1/C$, where C is the constant that bounds $f(x)/g(x)$
- In other words, the expected number of trials to get one acceptance is $1/p=C$

Efficiency

- If $g(x)$ is close to $f(x)$, then C is small, so p is large, and expected number of trials is small
- If $g \equiv f$, then $C=1$, so that $p=1$ and we finish the random variable generation always in one trial

How to Choose Proposal

- Before we even consider efficiency, we have to make sure the basic requirement of $g(x)$ holds
- Requirement 1: $g(x)$ should have at least the same support as $f(x)$

How to Choose Proposal

- Requirement 2: $g(x)$ should have a heavier tail than $f(x)$ in the case that $f(x)$ has unbounded support

How to Choose Proposal

Most generally, we must be able to find C that is finite; otherwise, the A-R procedure fails

How to Choose Proposal

Can we use the following to generate $\text{Exp}(1)$, i.e., with density $f(x) = e^{-x}$ for $x > 0$ and 0 otherwise:

- Two-sided exponential, i.e., $g(x) = \frac{1}{2} e^{-|x|}$, $-\infty < x < \infty$
- $\text{Unif}(0,1)$, i.e., $g(x) = 1$ for $0 < x < 1$ and 0 otherwise

Can we use the following to generate $N(0,1)$, i.e., with density $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$, $-\infty < x < \infty$:

- $\text{Exp}(1)$, i.e., $g(x) = e^{-x}$ for $x > 0$ and 0 otherwise
- Density $g(x) = K e^{-|x|^3}$ for some $K > 0$